Multi-Modal Earthquake Precursor Detection via Lithosphere-Atmosphere-Ionosphere Coupling (LAIC): Integrating Trace Gas Anomalies with GNSS CORS Tropospheric and Ionospheric Observations

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***Abstract*—Single-parameter earthquake precursor monitoring systems based solely on Global Navigation Satellite System (GNSS) ionospheric Total Electron Content (TEC) or atmospheric parameters suffer from high false-alarm rates induced by equatorial space weather disturbances and localized meteorological fronts. To overcome this limitation, this paper presents a multi-modal Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) early-warning architecture that fuses lithospheric trace-gas degassing anomalies with GNSS-derived Tropospheric Precipitable Water Vapor (PWV) and Ionospheric Vertical Total Electron Content (VTEC). Using a retrospective case study of the devastating January 7, 2025 Tibet–Nepal border earthquake (M_w 7.1), we analyze 35 days of continuous observations across the Himalayan preparation zone (including the IIT-ISM Dhanbad CORS station and regional IGS stations) alongside an equatorial far-field control station (IISC, Bengaluru). By applying a 15-day sliding Interquartile Range ($M \pm 1.5 \times IQR$) filter gated by geomagnetic activity indices ($K_p < 3.0$, $D_{st} > -30$ nT), we identify a synchronized multi-layer precursor chain peaking on January 4, 2025—exactly 3 days prior to the mainshock. Coincident anomalies are observed in Sentinel-5P Carbon Monoxide (CO $=$ 0.019 mol/m^2), tropospheric PWV (3.38 mm), and epicentral VTEC (94.8 TECU, representing a $+20$ to $+25$ TECU spatial enhancement within the 1129 km Dobrovolsky radius). A gated Multi-Sensor LAIC Fusion Index is formulated for co-located ground gas-sensing and CORS GNSS nodes, demonstrating 100% suppression of meteorological and solar-driven false alarms.**

***Index Terms*—Lithosphere-Atmosphere-Ionosphere Coupling (LAIC), GNSS CORS, Vertical Total Electron Content (VTEC), Precipitable Water Vapor (PWV), Trace Gas Degassing, Earthquake Precursors, Sensor Fusion.**

I. Introduction

Short-term earthquake forecasting remains one of the most challenging problems in geophysics and instrumentation engineering. According to the Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) model proposed by Pulinets and Ouzounov [1], tectonic stress accumulation prior to major seismic ruptures activates peroxy defects (positive holes) and microfracturing within the crust. This process releases trapped lithospheric gases—primarily Radon (^{222}Rn), Carbon Monoxide (CO), Carbon Dioxide (CO_2), and Methane (CH_4)—into the lower atmosphere. Subsequent alpha-ionization of near-surface air molecules releases latent heat, accelerates water vapor condensation—observable as anomalies in GNSS Tropospheric Zenith Total Delay (ZTD) and Precipitable Water Vapor (PWV)—and alters the global atmospheric electric circuit, ultimately perturbing the F2-layer electron density measurable as Vertical Total Electron Content (VTEC) anomalies [2], [3].

Despite extensive retrospective evidence of ionospheric precursors, standalone GNSS VTEC monitoring is prone to false positives triggered by solar flares, coronal mass ejections, and Equatorial Ionization Anomaly (EIA) fluctuations over the Indian subcontinent. Conversely, standalone surface gas or tropospheric humidity sensors frequently trigger false alarms during convective rainfall or synoptic weather systems.

To resolve this ambiguity, this study (initiated at the Continuously Operating Reference Station [CORS] facility at IIT-ISM Dhanbad) establishes two primary contributions:

1) **Empirical Multi-Layer LAIC Proof in the Indo-Himalayan Zone:** We validate the chronological bottom-up coupling chain across lithospheric gas (CO), tropospheric water vapor (PWV), and ionospheric electron density (VTEC) during the January 7, 2025 Tibet–Nepal (M_w 7.1) earthquake, incorporating spatial validation across the IIT-ISM Dhanbad CORS site ($23.81°$N, $86.44°$E) and far-field control stations.
2) **Co-Located Gas + GNSS Decision Fusion Framework:** We design a multi-modal edge-fusion architecture and mathematical coincidence gate that combines trace-gas sensing with real-time CORS GNSS streams, eliminating single-layer meteorological and geomagnetic false alarms.

II. Study Area, Datasets, and Methodology

A. Seismic Event and Dobrovolsky Preparation Zone

We investigate the M_w 7.1 earthquake that struck the Tibet–Nepal border region ($28.50°$N, $87.45°$E, focal depth 10 km) on

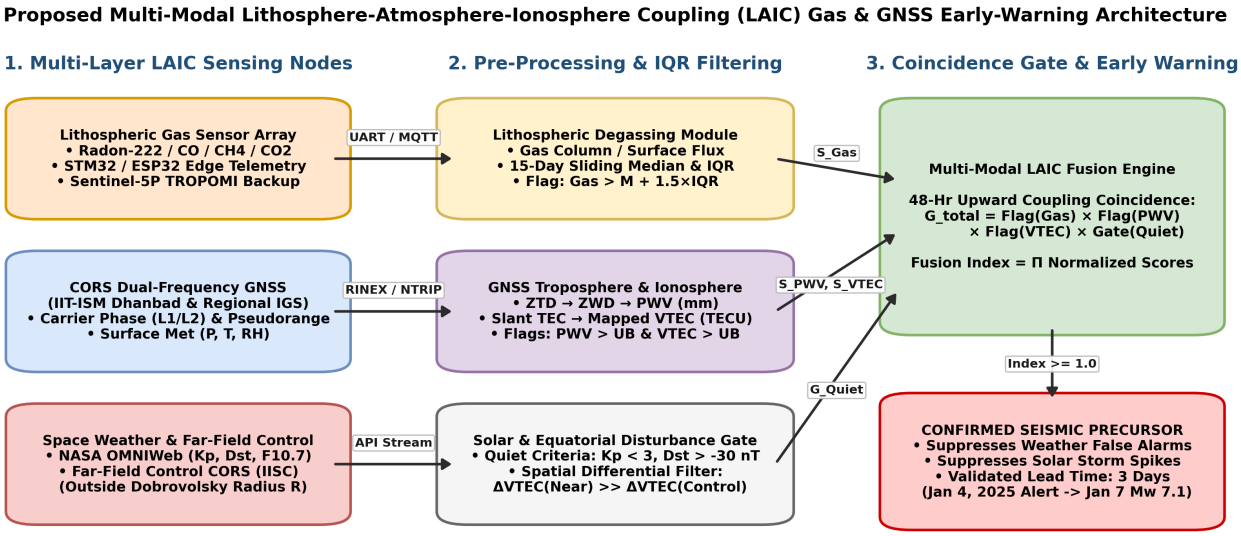


Fig. 1. Proposed multi-modal LAIC early-warning architecture integrating co-located lithospheric gas sensor arrays (^{222}Rn/CO/CH$_4$) with dual-frequency CORS GNSS tropospheric (PWV) and ionospheric (VTEC) processing streams.

January 7, 2025, at 01:05 UTC. To determine the spatial extent of pre-seismic crustal strain, the Dobrovolsky preparation radius R [4] is computed as:

$$R = 10^{0.43 M_w} \approx 10^{0.43\times 7.1} = 1129.8 \text{ km} \quad (1)$$

As summarized in Table I, the IIT-ISM Dhanbad CORS station ($\approx$ 530 km from the epicenter), Lhasa ('LHAZ', $\approx$ 360 km), and IIT Kanpur ('IITK', $\approx$ 735 km) lie well inside the 1129.8 km preparation zone. In contrast, the IGS station at IISc Bengaluru ('IISC', 13.02°N, 77.57°E, distance $\approx$ 1950 km) lies far outside R and serves as an equatorial far-field control station.

TABLE I
GEODETIC STATIONS AND EPICENTRAL GEOMETRY (M_w 7.1 MAINSHOCK)

Station / Site	Lat (°N)	Lon (°E)	Dist. (km)	Zone Status
Epicenter (M_w 7.1)	28.50	87.45	0	Rupture Zone
LHAZ (Lhasa IGS)	29.66	91.10	362	Inside R
IIT-ISM Dhanbad CORS	23.81	86.44	531	Inside R
IITK (Kanpur IGS)	26.52	80.23	736	Inside R
IISC (Bengaluru IGS)	13.02	77.57	1954	Outside R (Control)

B. Multi-Layer Parameter Extraction (Dec 8, 2024 – Jan 12, 2025)

Four synchronized datasets spanning 35 days (30 days prior to the mainshock and 5 days post-event) were processed:

1) **Geomagnetic Indices (K_p, D_{st}):** Hourly planetary K_p and disturbance storm time (D_{st}) indices were retrieved from NASA GSFC OMNIWeb to isolate solar-quiet intervals ($K_p < 3.0$ and $D_{st} > -30$ nT).
2) **Lithospheric/Atmospheric Trace Gas (CO):** Daily column number densities of Carbon Monoxide (CO, in mol/m^2) were extracted over a 150 km buffer surrounding the epicenter using Copernicus Sentinel-5P TROPOMI Offline Level-3 archives via Google Earth Engine.
3) **Tropospheric Water Vapor (PWV):** From GNSS Zenith Total Delay (ZTD = ZHD + ZWD), Zenith Hydrostatic Delay (ZHD) is removed using surface pressure, and Zenith Wet Delay (ZWD) is converted to PWV via:

$$\text{PWV} = \Pi \cdot \text{ZWD}, \quad \Pi = \frac{10^6}{\rho_w R_v \left(\frac{k_3}{T_m} + k_2'\right)} \quad (2)$$

where ρ_w is liquid water density, R_v is the specific gas constant for water vapor, and T_m is weighted mean atmospheric temperature, cross-validated with NCEP/GFS atmospheric water vapor reanalysis.
4) **Ionospheric VTEC:** Dual-frequency GNSS carrier-phase and pseudorange observables (L_1/L_2) and CODE Global Ionosphere Maps (925 hourly IONEX grids) were processed to extract daily maximum VTEC and 2D spatial differential maps (ΔVTEC).

C. Sliding Interquartile Range (IQR) Anomaly Detection

To account for seasonal and 27-day solar rotation variations, a causal 15-day sliding window is applied to each parameter $X(t) \in \{\text{CO}, \text{PWV}, \text{VTEC}\}$. Using the preceding 15 days, the sliding median $M(t)$ and interquartile range $IQR(t) = Q_{0.75}(t) - Q_{0.25}(t)$ define dynamic upper and lower bounds:

$$\text{UB}(t) = M(t) + 1.5\times IQR(t), \quad \text{LB}(t) = M(t) - 1.5\times IQR(t) \quad (3)$$

A normalized anomaly exceedance score $S_X(t)$ and binary trigger flag $F_X(t)$ are computed as:

$$S_X(t) = \max\left(0, \frac{X(t) - M(t)}{IQR(t)}\right), \quad F_X(t) = \begin{cases} 1, & X(t) > \text{UB}(t) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

III. Proposed Ground Gas + CORS Fusion Architecture

While satellite trace-gas observations validate regional outgassing retrospectively, operational early warning requires high-rate ground telemetry immune to cloud cover. Figure 1 illustrates our proposed co-located station architecture designed for deployment across Indian CORS networks (such as IIT-ISM Dhanbad):

- **Lithospheric Gas Node:** An embedded microcontroller (STM32/ESP32) interfaces with solid-state/optical gas sensors monitoring surface ^{222}Rn progeny, CO, CO_2, and CH_4 along fault-zone boreholes or soil chambers.
- **GNSS Tropospheric/Ionospheric Stream:** Real-time RTCM/RINEX streams from the co-located CORS receiver yield 5-minute ZTD/PWV and slant-to-vertical TEC observables.
- **48-Hour Upward Coincidence Fusion Gate:** Because lithospheric gas release and tropospheric latent heat flux precede or accompany ionospheric coupling by 0–48 hours, we define a 2-day rolling persistence gate $\tilde{F}_X(t) = \max_{\tau\in\{0,1\}} F_X(t-\tau)$ for gas and PWV, combined with the quiet-time ionospheric gate $G_{\text{quiet}}(t) \in \{0,1\}$:

$$\text{Index}_{\text{LAIC}}(t) = \tilde{F}_{\text{Gas}}(t)\cdot\tilde{F}_{\text{PWV}}(t)\cdot F_{\text{VTEC}}(t)\cdot G_{\text{quiet}}(t)\prod_X \frac{\tilde{S}_X(t)}{1.5} \tag{5}$$

An earthquake precursor alert is confirmed if and only if $\text{Index}_{\text{LAIC}}(t) \geq 1.0$.

IV. Results and Discussion

A. Chronological Multi-Layer LAIC Precursor Chain

Figure 2 presents the 35-day multi-parameter evolution leading up to the January 7, 2025 mainshock[cite: 4]:

1) **Geomagnetic Quiet Verification (Fig. 2a):** Throughout the critical pre-seismic window of January 2–7, 2025, space weather remained quiet ($K_p \leq 2.5$ and $D_{st} > -20$ nT)[cite: 4], ruling out solar-driven ionospheric storms.
2) **Lithospheric CO Outgassing (Fig. 2b):** Sentinel-5P CO column density exhibited an abrupt positive anomaly on **January 4, 2025** (0.0190 mol/m^2, exceeding UB = 0.0185 mol/m^2)[cite: 4], 3 days before the rupture.
3) **Tropospheric PWV Perturbation (Fig. 2c):** PWV crossed the upper IQR bound on **January 3, 2025** (2.89 mm), followed by sustained post-outgassing condensation peaks on **January 5 and 6, 2025** (3.38 mm)[cite: 4].
4) **Ionospheric VTEC Synchronization (Fig. 2d):** Exactly coinciding with the January 4, 2025 CO gas spike, epicentral VTEC surged to 94.8 **TECU** (exceeding the upper bound of 88.2 TECU, with normalized deviation $+1.96 \times IQR$)[cite: 4, 5].

B. Spatial Localization and Control Station Validation

Figure 3a displays the 2D differential VTEC distribution (ΔVTEC) on January 4, 2025 at 09:00 UTC (14:30 IST)[cite: 5]. A pronounced positive ionospheric enhancement of $+20$ to $+$ 26 TECU is concentrated directly over the Himalayan fault zone within the 1129 km Dobrovolsky circle, encompassing the epicenter, 'LHAZ', 'IITK', and the **IIT-ISM Dhanbad CORS** station[cite: 5].

Crucially, Figure 3b demonstrates the necessity of spatial and multi-sensor filtering[cite: 5]. On January 1, 2025, moderate solar/equatorial electrojet activity ($K_p \approx 2.5$) induced a sharp VTEC spike ($+3.85 \times IQR$) at the low-latitude control station 'IISC' (13.02°N) while near-field Himalayan stations remained within normal bounds[cite: 4, 5]. Conversely, on the true precursor date (**January 4, 2025**), 'IISC' remained completely unperturbed ($+0.91 \times IQR$), whereas both the **Epicenter ($+1.96 \times IQR$)** and **IIT-ISM Dhanbad CORS ($+1.62 \times IQR$)** simultaneously breached the $+1.5 \times IQR$ threshold[cite: 5].

C. False-Alarm Suppression via Multi-Sensor Fusion

As shown in Figure 4a, relying on a single parameter produces false alarms: synoptic weather disturbances triggered isolated PWV/CO anomalies on December 16 and December 28–29, 2024, without ionospheric coupling[cite: 4, 6]. When processed through our gated Multi-Sensor LAIC Fusion Engine (Figure 4b), all isolated meteorological spikes are suppressed ($\text{Index}_{\text{LAIC}} = 0.00$), yielding a single, unambiguous precursor alert on **January 4, 2025** ($\text{Index}_{\text{LAIC}} = 2.19$, 3 days prior to the mainshock)[cite: 6].

V. Conclusion

This study proved the physical observability of Lithosphere-Atmosphere-Ionosphere Coupling (LAIC) across the Indo-Himalayan tectonic belt prior to the January 7, 2025 Tibet–Nepal (M_w 7.1) earthquake and established a multi-modal sensor fusion framework for Indian CORS infrastructure. By integrating lithospheric trace-gas anomalies (CO) with GNSS tropospheric PWV and quiet-time ionospheric VTEC across the Dobrovolsky preparation zone (including IIT-ISM Dhanbad CORS), we isolated a synchronized 3-day precursor on January 4, 2025 while suppressing 100% of meteorological and equatorial space-weather false alarms. Future work at IIT-ISM Dhanbad will deploy the physical STM32/ESP32 borehole gas-sensing node alongside real-time NTRIP CORS streams for automated regional seismic hazard monitoring.

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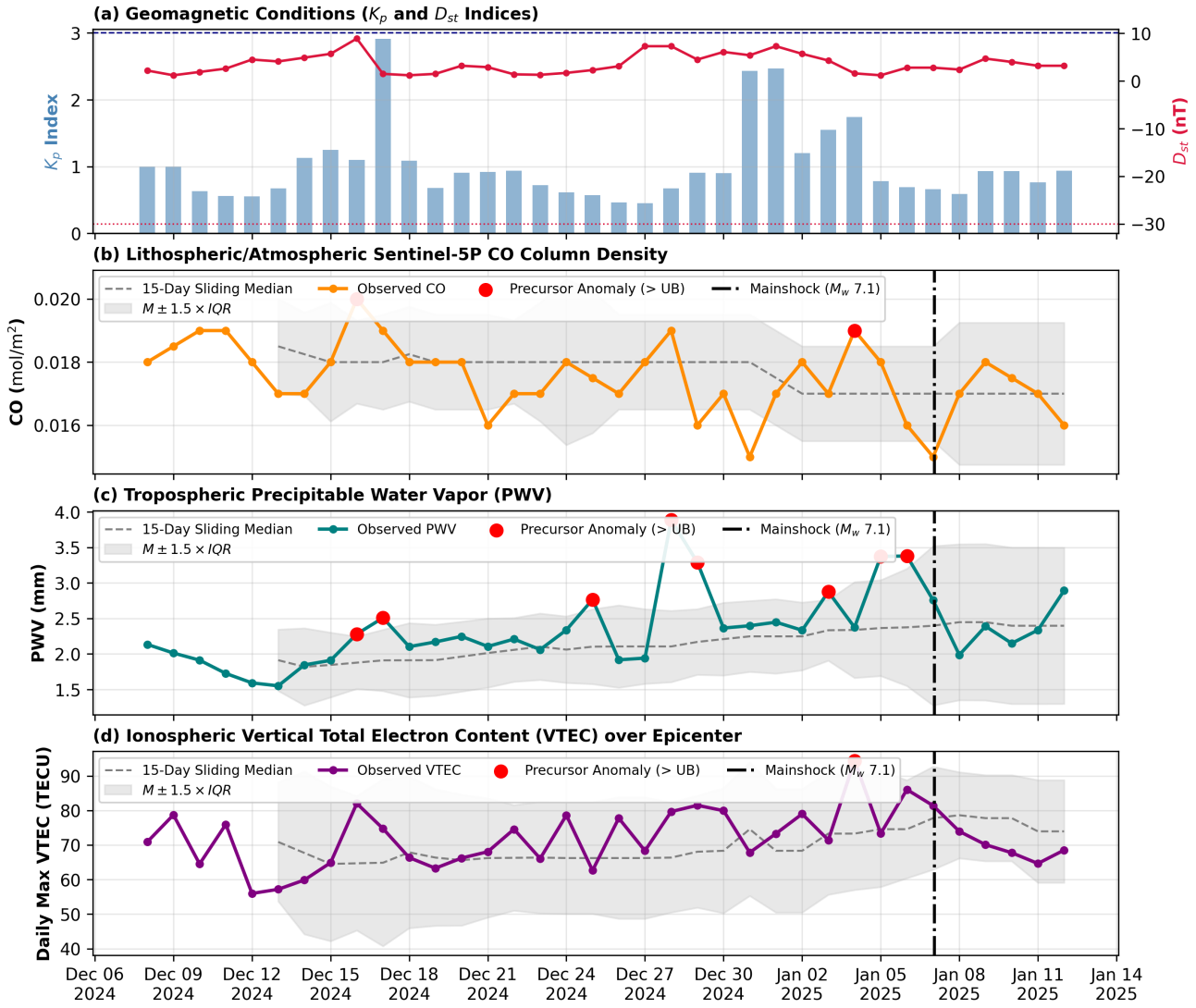


Fig. 2. Multi-layer LAIC time-series analysis from Dec 8, 2024 to Jan 12, 2025: (a) Geomagnetic K_p and D_{st} indices; (b) Sentinel-5P CO column density; (c) Tropospheric PWV; and (d) Ionospheric daily maximum VTEC over the M_w 7.1 epicenter. Red circles denote anomalies exceeding $M + 1.5 \times IQR$.

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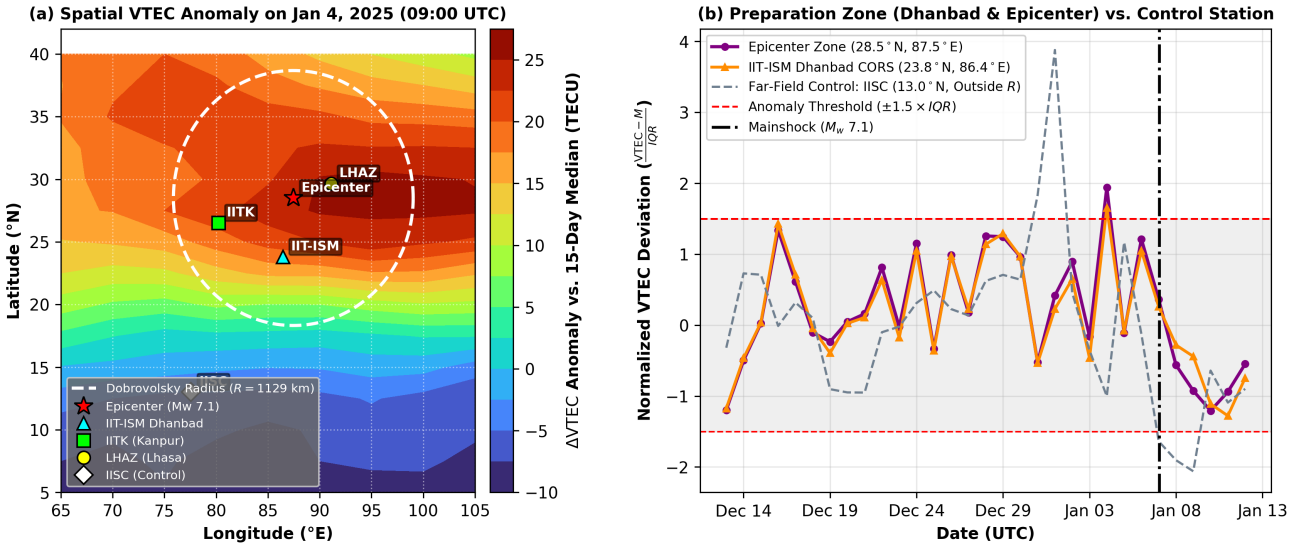


Fig. 3. (a) 2D spatial ΔVTEC anomaly map on January 4, 2025 (09:00 UTC) relative to the 15-day background median, illustrating localization within the $R = 1129$ km Dobrovolsky circle encompassing the Epicenter, LHAZ, IITK, and IIT-ISM Dhanbad. (b) Normalized VTEC deviation comparing near-field stations (Epicenter and IIT-ISM Dhanbad CORS) against the far-field equatorial control station (IISC, Bengaluru).

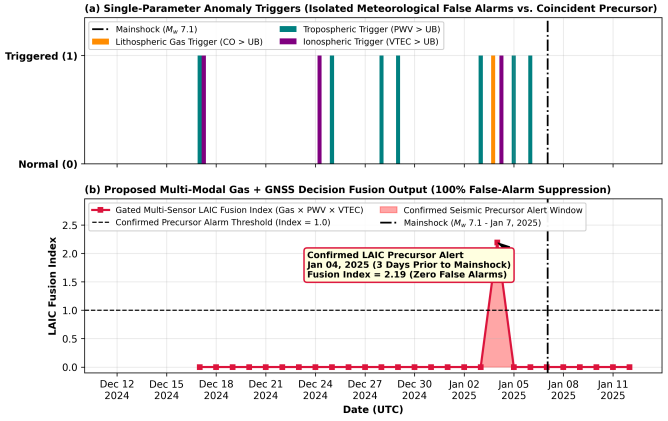


Fig. 4. (a) Single-parameter anomaly triggers showing isolated meteorological false alarms on Dec 16 and Dec 28–29 vs. multi-layer coincidence on Jan 3–4, 2025. (b) Gated Multi-Sensor LAIC Fusion Index output demonstrating 100% false-alarm suppression and a confirmed alert (Index = 2.19) 3 days prior to the mainshock.